

EPPLEY FIELD, NE, USA

WMO: 725500

Lat: 41.310N

Lon: 95.899W

Elev: 982

StdP: 14.18

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14942

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.8	(c) 1.6	(d) -10.9	(e) 3.2	(f) -1.4	(g) -6.7	(h) 4.0	(i) 2.6	(j) 31.9	(k) 21.2	(l) 28.2	(m) 20.7	(n) 9.0	(o) 350	(p) 0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	95.0	76.1	92.1	75.3	89.2	73.9	79.5	90.0	77.7	88.0	76.1	85.8	14.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.4	143.0	86.5	74.6	134.5	84.5	73.0	126.9	82.6	43.8	90.0	41.9	87.9	40.2	85.9	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 26.8	(b) 24.2	(c) 20.8	(e) -10.3	(f) 99.8	(g) 4.7	(h) 3.6	(i) -13.7	(j) 102.4	(k) -16.5	(l) 104.5	(m) -19.2	(n) 106.5	(o) -22.6	(p) 109.2		
(4) 26.8	(5) 24.2	(6) 20.8	(7) -10.8	(8) 82.1	(9) 4.7	(10) 1.4	(11) -14.2	(12) 83.1	(13) -16.9	(14) 83.9	(15) -19.6	(16) 84.7	(17) -22.9	(18) 85.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.1	23.8	28.0	40.3	52.2	63.1	73.3	77.6	75.1	67.5	54.2	40.3	28.0	
	DBStd	21.10	12.12	12.24	12.18	9.98	8.58	6.73	5.99	5.66	8.28	9.48	10.10	10.99	
	HDD50	2940	812	619	346	90	6	0	0	0	1	64	318	684	
	HDD65	5947	1277	1036	768	397	141	12	1	3	69	353	742	1148	
	CDD50	3701	0	3	45	157	412	700	857	780	525	194	27	1	
	CDD65	1233	0	0	2	14	82	263	393	318	143	18	0	0	
	CDH74	12325	0	0	30	197	850	2630	4197	2896	1339	180	6	0	
	CDH80	4937	0	0	4	54	281	1050	1884	1149	471	44	0	0	
Wind	WSAvg	10.0	10.3	10.6	11.3	11.9	10.4	9.6	8.5	8.1	9.2	9.8	10.2	9.9	
Precipitation	PrecAvg	33.1	0.9	1.1	2.0	3.4	4.9	5.2	3.9	4.7	2.5	2.1	1.3	1.0	
	PrecMax	43.1	1.7	2.5	4.5	7.4	9.2	11.3	9.6	10.3	6.7	5.2	2.7	4.8	
	PrecMin	22.3	0.3	0.3	0.2	0.5	1.7	0.7	0.3	1.3	0.4	0.0	0.0	0.0	
	PrecStd	5.6	0.4	0.6	1.1	1.5	2.1	2.7	2.2	2.6	1.7	1.5	0.9	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.7	65.5	77.9	86.7	92.4	95.9	101.0	97.2	93.3	85.7	73.2	58.8	
		MCWB	44.0	50.2	58.5	64.0	70.8	74.6	75.9	75.9	72.9	67.5	58.0	49.9	
	2%	DB	49.5	57.2	71.5	79.6	86.8	92.4	95.9	93.1	89.4	78.9	66.3	52.3	
		MCWB	41.3	47.4	56.6	61.5	68.8	74.4	77.4	76.1	72.4	63.9	55.0	45.0	
	5%	DB	44.3	50.5	65.2	74.3	82.7	89.2	92.8	89.4	85.5	74.1	60.7	47.5	
		MCWB	38.4	42.6	54.1	58.3	65.9	73.0	76.8	75.2	70.6	60.9	51.4	41.3	
	10%	DB	39.5	44.8	59.3	69.4	78.7	86.2	89.3	86.3	81.7	69.7	55.9	42.7	
		MCWB	35.4	39.1	50.0	56.0	64.6	71.8	75.4	73.5	68.7	58.8	48.0	37.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.8	54.9	63.2	66.9	74.2	79.4	82.2	80.9	76.8	71.1	60.8	53.3
			MCDB	53.9	60.4	72.5	80.8	85.4	89.7	92.6	91.2	88.9	81.0	68.4	57.1
2%		WB	42.0	47.9	59.5	63.5	71.3	77.0	80.2	78.8	74.5	66.8	56.7	45.9	
		MCDB	49.1	56.5	66.6	75.9	83.1	88.3	91.0	88.6	84.9	74.5	63.9	50.7	
5%		WB	38.7	42.9	55.1	60.4	68.8	75.2	78.3	77.0	72.7	63.7	52.6	41.6	
		MCDB	43.6	49.8	64.1	71.7	79.3	86.0	88.7	86.4	82.3	71.2	59.9	47.0	
10%		WB	35.5	39.3	50.2	57.4	66.2	73.4	76.8	75.2	70.8	60.2	48.3	38.2	
		MCDB	39.4	44.5	59.5	67.6	75.9	83.9	86.7	83.7	79.5	68.1	55.4	42.6	
Mean Daily Temperature Range	5% DB	MDBR	17.5	18.2	20.9	21.7	20.7	19.7	19.0	19.0	21.6	21.0	19.2	16.6	
		MCDBR	24.9	26.9	30.2	30.0	25.9	22.2	22.0	22.1	24.5	26.7	27.1	24.0	
		MCWBR	17.7	17.7	17.3	15.8	11.6	9.6	9.0	9.5	11.0	13.9	16.4	17.2	
	5% WB	MCDBR	23.3	25.6	26.2	25.5	22.1	19.8	19.4	19.5	20.9	22.3	24.4	21.9	
		MCWBR	17.1	17.5	15.8	15.1	11.2	10.4	9.8	9.9	10.9	13.7	16.4	16.4	
Clear-Sky Solar Irradiance	taub	0.283	0.318	0.332	0.388	0.412	0.417	0.429	0.418	0.378	0.335	0.300	0.286		
	taud	2.445	2.365	2.390	2.253	2.255	2.300	2.260	2.299	2.366	2.473	2.498	2.450		
	Ebn at Noon	279	282	291	280	275	273	268	268	272	273	270	267		
	Edh at Noon	24	31	34	42	43	41	42	40	35	28	23	23		
All-Sky Solar Radiation	RadAvg	633	934	1209	1545	1837	2040	2058	1776	1466	1023	711	533		
	RadStd	57	56	109	110	139	160	135	116	95	120	49	53		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+202	+126
(47) Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page